



Project Name: AI-Powered Employee Expense Reimbursement Management System

Objective

Develop a simple, user-friendly web-based expense reimbursement platform that allows employees to submit reimbursement requests with minimal effort. The system should automate notifications, document validation, approvals, record management, reporting, and AI-assisted bill processing.

The primary goal is to reduce manual intervention, improve transparency, accelerate reimbursements, and maintain complete audit records.

1. Business Requirements

Employee Requirements

Employees should be able to:

Expense Submission

- Enter reimbursement amount
- Select expense category
- Add expense date
- Add expense description
- Upload bills/invoices/receipts
- Submit request

Supported Categories

Examples:

- Travel
- Food & Meals
- Internet Reimbursement
- Mobile Bill
- Office Supplies
- Client Meeting Expenses
- Training Expenses





- Other

Employee Dashboard

Employees should be able to view:

- Submitted claims
- Pending claims
- Approved claims
- Rejected claims
- Reimbursement history
- Payment status

Finance Team Requirements

Finance team should be able to:

Review Claims

View:

- Employee details
- Amount claimed
- Receipt details
- Expense category
- OCR extracted information

Actions:

- Approve
- Reject
- Request Clarification
- Mark as Paid

HR Team Requirements

HR should receive visibility into:





- New reimbursement submissions
- Employee reimbursement trends
- Policy violations
- High-value claims
- Pending approvals

HR should not necessarily approve every claim unless configured.

Admin Requirements

Admin should have full control.

Admin Capabilities

- User management
- Role management
- Expense policy management
- Approval workflow configuration
- Report generation
- Data export
- Audit log review
- AI model configuration

2. AI Features

OCR Bill Scanning

When a receipt is uploaded:

AI automatically extracts:

- Vendor Name
- Invoice Number
- Invoice Date
- GST Number (if available)
- Amount
- Tax Amount





- Total Amount

Receipt Validation

AI should detect:

- Duplicate bills
- Manipulated receipts
- Blurry images
- Missing information
- Invalid dates
- Future dates

Policy Validation

AI compares claim against company policies.

Examples:

- Meal limit ₹500/day
- Internet reimbursement ₹1000/month
- Travel cap ₹5000/trip

If exceeded:

- Flag claim
- Notify Finance
- Notify HR

Smart Recommendations

AI should automatically suggest:

- Category classification
- Policy compliance status
- Approval recommendation

Examples:

- Low Risk
- Medium Risk





- High Risk

3. Notification System

Email Notifications

Employee

Receive emails when:

- Claim submitted
- Claim approved
- Claim rejected
- Additional information requested
- Payment completed

Finance Team

Receive immediate email:

Subject:

"New Reimbursement Request Submitted"

Details:

- Employee Name
- Employee ID
- Amount
- Category
- Submission Date

HR Team

Receive heads-up notification:

Subject:

"Employee Reimbursement Request Alert"





Details:

- Employee Name
- Department
- Claimed Amount
- Category

Admin

Receive alerts for:

- Large claims
- System errors
- Fraud detection flags

4. Approval Workflow

Standard Workflow

Employee



Finance Review



HR Notification



Approval/Rejection



Payment Processing



Employee Notification

Advanced Workflow

Employee



AI Validation





Finance Review



Manager Approval (Optional)



HR Visibility



Payment Confirmation



Closure

5. Database Requirements

Employee Master

Store:

- Employee ID
- Name
- Email
- Department
- Designation
- Manager

Expense Records

Store:

- Claim ID
- Employee ID
- Amount
- Category
- Description
- Submission Date
- Approval Status
- Payment Status

Receipt Repository





Store:

- Original Receipt
- OCR Data
- Validation Results
- Upload Timestamp

Audit Logs

Track:

- User Actions
- Approval Changes
- Rejections
- Edits
- Downloads

6. Reporting Module

Admin Reports

Generate reports for:

Monthly Expense Summary

- Total claims
- Total reimbursements
- Department-wise spending

Employee Expense Report

- Employee-wise history

Category Report

- Travel expenses
- Meal expenses
- Internet expenses

Rejected Claims Report



- Rejection reasons
- Frequent violations

7. Search & Retrieval

Admin should retrieve data using:

- Employee Name
- Employee ID
- Claim Number
- Date Range
- Category
- Amount Range
- Status

8. User Interface Requirements

Employee Portal

Keep extremely simple.

Screen 1

Submit Expense

Fields:

- Amount
- Category
- Date
- Description
- Upload Receipt

Button:

Submit

Screen 2

My Claims

Columns:





- Claim ID
- Amount
- Status
- Submitted Date

Finance Dashboard

Cards:

- Pending Approvals
- Approved Claims
- Rejected Claims
- Paid Claims

Admin Dashboard

Cards:

- Total Claims
- Total Reimbursements
- Fraud Alerts
- Pending Reviews
- Monthly Spend

9. Security Requirements

Authentication

- Microsoft SSO
- Google SSO
- Email OTP

Authorization

Role-based access:

- Employee
- Finance
- HR





- Manager
- Admin

Data Security

- Encrypted storage
- HTTPS
- Audit logging
- Backup system

10. Technology Stack Recommendation

Frontend

- React.js
- Next.js
- Tailwind CSS

Backend

- Node.js
- NestJS

or

- Python FastAPI

Database

- PostgreSQL

Storage

- AWS S3
- Azure Blob Storage

AI Services

- Azure Form Recognizer
- AWS Textract
- Google Document AI



- OpenAI Vision for receipt extraction

Notifications

- SendGrid
- Microsoft Graph API
- SMTP

